

A pre-feasibility study for the Production of AlJamid and dairy products.



2017

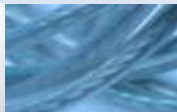


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1. Executive summary

يهدف المشروع الى تصنيع الجميد ومشتقات الألبان في محافظة الكرك. وسيقوم المشروع المقترح باستهداف المراكز التجارية، والسوبرماركت، بالإضافة إلى المطاعم في مختلف أنحاء المملكة بشكل رئيسي مع وجود فرصة للتوسع والتصدير للأسواق العربية وتغطية جزء من طلب الأقليات العربية المتزايد في الأسواق الأوروبية.

The project aims to manufacture Al-Jamid and dairy products in Karak governorate. The proposed project will target commercial centers, supermarkets and restaurants throughout the Kingdom, with an opportunity to expand and export to Arabian markets and cover the growing demand of Arabian minorities in European markets.

محلياً، هناك طلب مرتفع على المنتجات المصنعة من حليب الأغنام. ويشكل حليب الأغنام حوالي ٢٤٪ من كامل إنتاج الحليب في المملكة. يشكل التصنيع على نطاق صغير جزءاً مهماً من سبل معيشة الأسر التي تعتمد على الأغنام والماعز. ويعتبر الجميد المصنع من حليب الأغنام بشكل رئيسي أحد أهم المنتجات المصنعة والمستهلكة في المنطقة من قديم الزمان.

Locally, there is a high demand for products made of sheep's milk. Sheep's milk is about 24% of the entire production of milk in the Kingdom. Small-scale manufacturing is an important part of the livelihoods of families that depend on sheep and goats. Jamid, is made of sheep's milk, is one of the most important manufactured and consumed products in the governorates.

Table 1 economical indicators

Indicator	
Total Investment Cost (Jordan Dinars)	402,373
Internal Rate of Return (IRR)	45.5%
Pay-Back Period (Years)	2
Net Present Value (Jordan Dinars)	2,146,000
Job Opportunities provided by the Project	18

Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

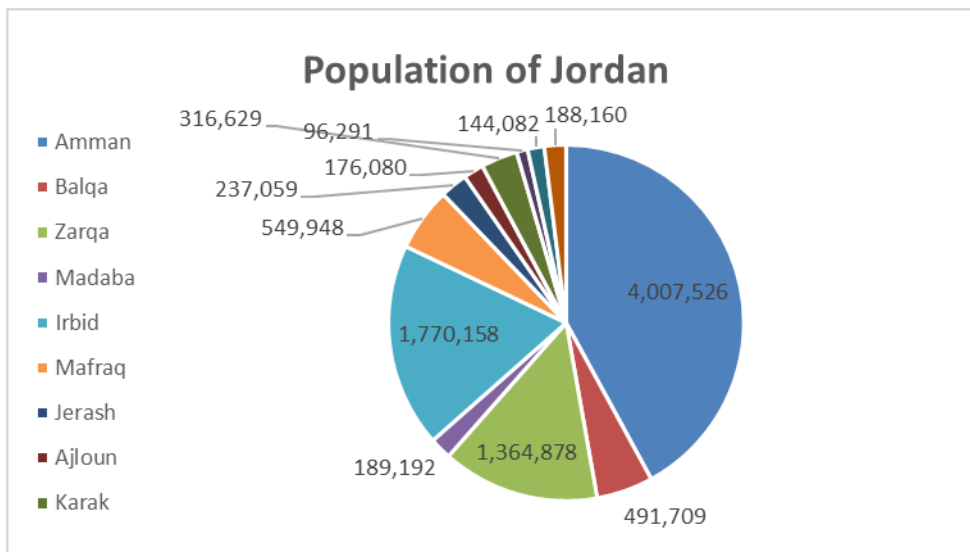
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafrq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table 2 Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

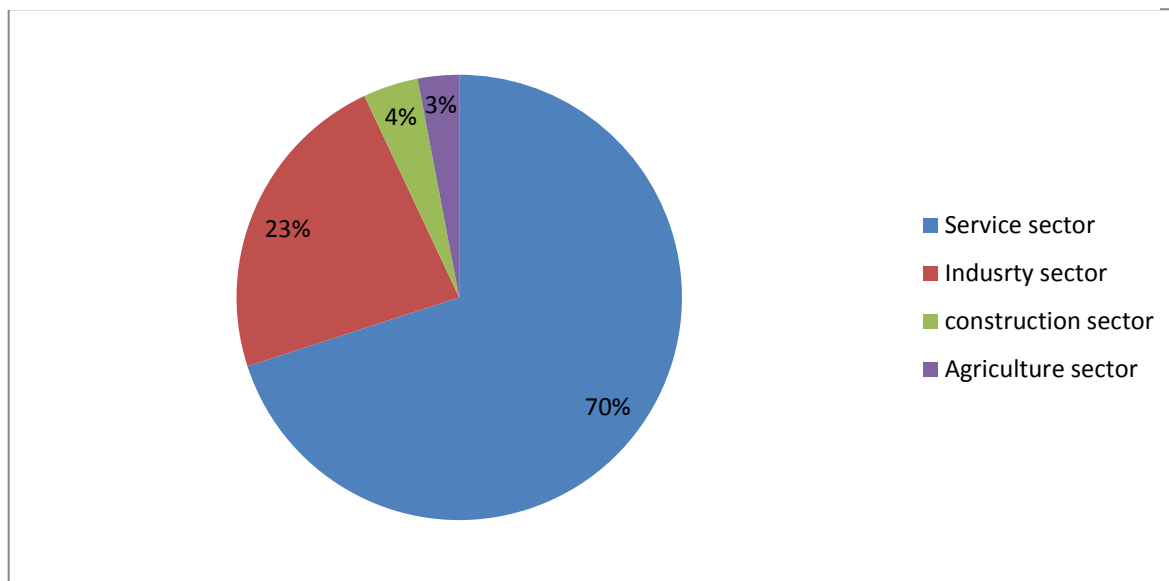
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure 2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and

medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance

of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

Furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of

capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

2. Market analysis

Project description

يقوم المشروع على فكرة تلخص في إقامة مصنع لتصنيع الجميد الكري بشكل رئيسي وانتاج مشتقات الألبان كالزبدة والسمنة البلدية التي تكون مصاحبة لعملية انتاج الجميد، بطاقة انتاجية تقدر بحوالي 100 طن من الجميد. حيث أن مثل هذه المشاريع تعد من المشاريع الزراعية والمرتبطة ارتباطا مباشرا بالتصنيع والتسويق.

The project is based on the idea of establishing a factory for the manufacturing Aljamid and the production of dairy products such as butter and ghee which are associated with the process of producing Aljamid, with a capacity estimated at 100 tons of Aljamid. As such projects are considered agricultural projects directly related to manufacturing and marketing.

أن مثل هذه المشاريع تعد من المشاريع التي يمكن بكل سهولة التحكم في حجم المشروع فيها، حيث أن معظم الآلات ومعدات المصنع يمكن أن تتوفر محليا بالشكل والحجم المطلوب. لذا ويهدف إيجاد أكبر قدر من فرص العمل فان الحجم المقترح لهذه المصانع يتراوح ما بين 1.5-2 طن في اليوم. ويمكن اعتبارها من المشاريع الريادية التي تحتاج إلى تدريب الكوادر والعمال قبل البدء بتنفيذ المشروع. ويمكن أن يؤدي هذا المشروع إلى فتح آفاق جديدة في تشغيل الأيدي العاملة.

Such projects are considered of the projects that can easily control the size of the project, since most of the required machines and equipment can be available locally. Therefore, in order to create the largest number of jobs, the production of these factories ranges between 1.5-2 tons per day. It can be considered as one of the pioneer projects that need to train its workers before starting the project. This project may lead to open new horizons in employment.

Project objectives

يحتاج السوق الأردني بشكل كبير لهذه المادة اذ بلغ معدل قيمة المستوردات من الجميد للأعوام (2014-2015) حوالي 3136 طن بقيمة تصل الى 7.9 مليون دينار سنويا. ويهدف المشروع إلى:

1. تحقيق عائد مالي مجدي لمالك المشروع،
2. تصنيع وإنتاج جميد كركي بنوعية مميزة وجيدة،
3. تصنيع الزبدة والسمنة ذات النوعية الجيدة .
4. خلق فرص عمل في المناطق الأقل نمواً، وتحسين الأحوال الاقتصادية والاجتماعية لمربي الماشية من خلال شراء الحليب الطازج
5. تصدير الجميد الكركي الى الاسواق العربية والاجنبية يحمل علامة تجارية مميزة.

The Jordanian market needs a great deal of this material as the average value of imports from AlJamid for the years (2014-2015) is about 3136 tons worth 7.9 million JD per year. The project aims to:

1. Achieving a profitable return to the owner of the project,
2. Manufacturing and producing good quality of AlJamid AlKarak,
3. Manufacturing good quality of butter and ghee.
4. Providing job opportunities and improving the economic and social conditions of the breeders by purchasing fresh milk
5. Exporting AlJamid AlKarak to Arabian and foreign markets.

Proposed product

اشتهرت الكرك بصناعة الجميد الذي ينتج غالبا في المنازل من حليب النعاج أو حليب الماعز ويتميز الجميد الكركي بنوعية عالية ومذاق طيب، لأن النعاج والماعز في الكرك تتغذى على أعشاب طبيعية متوفرة بكثرة في الكرك، فيما يوجد بعضها في مراعي الكرك فقط ومنها القرطم البان والشيخ.

AlKarak governorate is known for producing al-Jamid, which is often produced in homes from goat's milk or ewes's milk and are characterized by their high quality and good taste. The ewes and goats in AlKarak governorate feed on natural herbs, which are widely available in AlKarak.

يقوم المشروع على فكرة تلخص في إنشاء معمل ألبان يكون متخصص في معالجة حليب الأغنام وتحويله إلى منتجات ألبان أهمها الجميد والزبدة والسمن البلدي من أجل سد جزء بسيط من العجز في الميزان التجاري في منتجات الألبان وخاصة الجميد ويمكن أن ينتج الجبنة البيضاء.

The project is based on the idea of establishing a dairy factory specialized in the processing of sheep's milk and turning it into dairy products, the most important of which is AlJamid, butter and margarine. The plant repaid a fraction of the deficit in trade balance of dairy products, especially AlJamid in addition to producing white cheese.

إن المشروع المقترح عبارة عن مصنع البان لتصنيع مشتقات الألبان وأهمها الجميد بالإضافة إلى الزبدة والسمنة في أحد المواقع الرئيسية من المناطق القريبة من مصادر الحليب في محافظة الكرك.

The proposed project is a dairy plant for manufacturing dairy products, especially AlJamid in addition to butter and margarine at one of the main sites near the sources of milk in Al Karak Governorate.

Market size

تشير بيانات وزارة الزراعة الى ان الكميات المستوردة من الجميد المخصص للاستهلاك في اكياس مفرغة من الهواء قد بلغ 2680 طن، تم استيرادها من سوريا ومصر وتركيا، وهناك نوع آخر من الجميد والمخصص للصناعة قد بلغ كمية المستوردات منه حوالي (600) طن. ويبين الجدول التالي قيمة وكمية المستوردات من الجميد خلال الفترة الزمنية (2011-2015)، حيث بلغ متوسط كمية المستوردات من الجميد لوحده (2,379) طن بقيمة تقدر بحوالي (5,513) الف دينار، وتبين من خلال الجدول أن هناك عجزا في الميزان التجاري لمادة الجميد لسنة 2014 قد بلغ حوالي (-7,950) الف دينار.

Data from the Ministry of Agriculture indicates that the quantities imported from AlJamid allocated for consumption amounted to 2680 tons, which were imported from Syria, Egypt and Turkey, and there is another type of AlJamid dedicated for the industry, which amounted at about 600 tons. The following table shows the value and quantity of imports of AlJamid during the period (2011-2015). The average quantity of imports of Al Jamid alone was 2,379 tons with a value estimated at (5,513) thousand JD. The table shows that there is a deficit in trade balance of Al Jameed for the year 2014 that has reached about (-7,950) thousand JD.

وبين (جدول 4 و جدول 5) الكميات المستوردة من الزبدة والسمن البلدي والتي تكون مصاحبة لانتاج الجميد. حيث يتم معالجة الحليب وينتج مصل الحليب لانتاج الجميد وينتج الدهن لانتاج الزبدة والسمن البلدي. وتشير الأرقام كذلك الى ان هناك كميات متزايدة من الزبدة والسمن البلدي يتم استيرادها سنويا بقيمة تبلغ في المتوسط (5.8) مليون دينار للزبدة البلدية و(4.7) مليون دينار للسمن البلدي.

Table (4 and 5) shows the quantities imported from butter and ghee that are associated with the production of AlJameed. Where the milk is processed and produces whey to produce Al-Jamid and fat is produced to produce butter and margarine. The figures also indicate that there are increasing quantities of butter and margarine are imported annually with an average value of (5.8) million JD for butter and (4.7) million JD for margarine.

Table 3 Values and quantities of imports, exports, re-exports and trade deficit of AlJmeed

year	Value of imports (Thous.dinars)	Amount of imports (ton)	Value of national exports (Thous.dinar)	Amount of national exports (ton)	Value of re-exports (Thous.dinar)	Amount of re-exports (ton)	Total value of exports (Thous.dinar)	Value of trade deficit (Thous.dinar)
2011	3,494	1,803	568	254	285	137	853	-2,641
2012	1,765	874	1,177	427	310	141	1,488	-277
2013	6,538	2,947	1,090	402	224	225	1,314	-5,225
2014	9,054	3,550	1,098	400	6	1	1,104	-7,950
2015	6,714	2,722	1,621	608	495	257	2,116	-4,597
Avg.	5,513	2,379	1,111	418	264	152	1,375	-4,138

Source: Department of statistics (2017) external trade bulletin

http://www.dos.gov.jo/dos_home_a/main/linked-html/external_trade.htm

Table 4 Values and quantities of imports, exports, re-exports and trade deficit of butter

year	Value of imports (Thous.dinar)	Amount of imports (ton)	Value of re-exports (Thous.dinar)	Amount of re-exports (ton)	Total value of exports (Thous.dinar)	Value of trade deficit (Thous.dinar)
2011	5,211	1,423				-5,211
2012	6,158	1,900	122	30	122	-6,037
2013	4,819	1,423	60	17	60	-4,758
2014	6,463	1,959	18	4	18	-6,445
2015	6,593	2,324	93	50	93	-6,500
Avg.	5,849	1,806	73	25	73	-5,790

Source: Department of statistics (2017) external trade bulletin

http://www.dos.gov.jo/dos_home_a/main/linked-html/external_trade.htm

Table 5 Values and quantities of imports, exports, re-exports and trade deficit of margarine

year	Value of imports (Thous.dinar)	Amount of imports (ton)	Value of national exports (Thous.dinar)	Amount of national exports (ton)	Value of re-exports (Thous.dinar)	Amount of re-exports (Ton)	Total value of exports (Thous.dinar)	Value of trade deficit (Thous.dinar)
2011	3,528	704	25	18			25	-3,502
2012	4,504	1,142	22	4			22	-4,481
2013	4,282	967			102	64	102	-4,180
2014	5,416	1,168	60	62	73	18	132	-5,284
2015	5,636	1,570			13	2	13	-5,624
Avg.	4,673	1,110	36	28	62	28	59	-4,614

Source: Department of statistics (2017) external trade bulletin

http://www.dos.gov.jo/dos_home_a/main/linked-html/external_trade.htm

تم توقع الطلب المحلي على الجميد استنادا الى دراسة نفقات الاسرة لعام 2013 الصادرة من دائرة الاحصاءات العامة والتي قدر فيها متوسط استهلاك الفرد بحوالي (480) غم سنويا. وهذا المتوسط يختلف حسب المحافظات (جدول 7). ومن خلال الاطلاع على نشرة التعداد السكاني لعام 2015 وبافتراض نسبة النمو السكاني 2.2% سنويا للفترة القادمة كما تشير بذلك دائرة الاحصاءات العامة، تم عمل اسقاط سكاني خلال الفترة (2018-2027) ومن ثم ضرب عدد السكان المتوقع بمتوسط استهلاك الفرد من الجميد (على فرض ثبات معدل الاستهلاك). تم تقدير الطلب المستقبلي على الجميد خلال الفترة (2018-2027) كما هو مبين في جدول 7.

Local demand for Al-Jamid was expected based on the study of household expenditure for 2013 issued by the Department of Statistics, in which average per capita consumption was 480 g per year. This average varies by governorates (Table 7). By viewing the population census of 2015 and assuming population growth rate of 2.2% per year for the up-coming period, as indicated by the Department of Statistics, a population projection was carried out during the period (2018-2027) and then the expected population was multiplied by the average per capita consumption of AlJamid (Assuming a constant rate of consumption). The future demand for Al Jamaid was estimated during the period (2018-2027) as shown in Table 7.

Table 6 Per capita consumption of AlJmeed and expected population and demand for 2017

Governorates	population 2017	Per capita consumption (kg/year)	Expected demand (tons)
Amman governorate	4,197,886	0.61	2,561
Balqa governorate	515,035	0.44	227
Zarqa governorate	1,429,690	0.27	386
Ma'daba governorate	198,139	1.17	232
Irbid governorate	1,854,168	0.12	223
Mafraq governorate	576,033	0.2	115
Jarash governorate	248,274	0.25	62
Ajloun governorate	184,456	0.21	39
Karak governorate	331,624	1.06	352
Tafilah governorate	100,897	1.28	129
Ma'an governorate	150,928	1.05	158
Aqaba governorate	197,094	0.82	162
Jordan	9,984,223	0.48	4,644

Source: Department of Statistics (2017), household expenses for 2013

<http://web.dos.gov.jo/sectors/economic/economy/economy-surveys/?lang=ar>

وذلك على افتراض انه لن يكون هناك زيادة على معدل الطلب على الجميد (أي لن يكون هناك تغير في النمط الاستهلاكي) وستكون الزيادة فقط في حدود الزيادة السنوية في النمو السكاني بنسبة 2.2 %. وهناك طلب إضافي على الجميد من قبل المطاعم قدرت بحوالي 500 طن سنوياً نتيجة زيادة عدد المطاعم التي تقدم خدمة المناسف الجاهزة في مناسبات الأفراح والأعراس. حيث انه من الصعب تحديد النسبة المئوية لطلب المطاعم نظراً لعدم توفر البيانات الدقيقة الخاصة بذلك، وبذلك يتوقع أن يكون حجم الطلب للسنوات 2018-2027 كما يبين جدول 7:

On the assumption that there will be no increase in demand for AlJamid (there will be no change in the consumption pattern) and the increase will only be within the annual increase in population growth by 2.2%. There is an additional demand for Al Jamid by restaurants about 500 tons per year as a result of the increase in the number of restaurants that offer Almanasf on different occasions. As it is difficult to determine the percentage of restaurant demand due to the lack of accurate data for this, and therefore the demand is expected to be for the years 2018-2027 as shown in Table 7:

Table 7 Estimate expected demand for Algemeed during the period (2018 – 2027), in ton

Governorates	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Amman governorate	2,561	2,617	2,675	2,733	2,794	2,855	2,918	2,982	3,048	3,115
Balqa governorate	227	232	237	242	247	253	258	264	270	276
Zarqa governorate	386	395	403	412	421	430	440	450	459	470
Ma'daba governorate	232	237	242	247	253	258	264	270	276	282
Irbid governorate	223	227	232	238	243	248	254	259	265	271
Mafraq governorate	115	118	120	123	126	128	131	134	137	140
Jarash governorate	62	63	65	66	68	69	71	72	74	75
Ajloun governorate	39	40	40	41	42	43	44	45	46	47
Karak governorate	352	359	367	375	383	392	401	409	418	428
Tafaileh governorate	129	132	135	138	141	144	147	150	154	157
Ma'an governorate	158	162	166	169	173	177	181	185	189	193
Aqaba governorate	162	165	169	173	176	180	184	188	192	197
Total demand of home	4,644	4,747	4,851	4,958	5,067	5,178	5,292	5,409	5,528	5,649
Restaurants	500	511	522	534	545	557	570	582	595	608
Total demand	5,144	5,258	5,373	5,491	5,612	5,736	5,862	5,991	6,123	6,257

Source: Team of the study

The analysis of competitors

بالرغم أنه لا توجد مشاكل هامة لتسويق منتجات الألبان المصنوعة من حليب الماعز والأغنام، حيث يتم تسويقها على صورة لبن ولبنة وجبنة وجميد، بحيث تكون الجبنة والجميد ضمن المنتجات الصالحة للتخزين لفترة طويلة، ويعتمد مربو الماشية في محافظة الكرك على إنتاج الجبنة المغلية والجميد وذلك لانتشار قطعانهم في أماكن متفرقة وبعبدة عن مناطق الاستهلاك الطازج للحليب واللبن واللبننة في العاصمة وما حولها. وتباع هذه المنتجات بأسعار مرتفعة نسبياً مقارنة مع المنتجات المستوردة لارتفاع جودة المنتجات منها وتفضيل المستهلك الانتاج المحلي على المستورد منها.

Although there are no significant problems for marketing of dairy products made from goat's milk and sheep, they are marketed in the form of yogurt, cheese, Labaneh and AlJamid, so that cheese and AlJamid are among the products that are suitable for long-term storage. The cattle farmers in AlKarak governorate rely on the production of boiled cheese and AlJamid for the spread of their livestock and away from the areas of fresh consumption of milk, yogurt and Labaneh in and around the capital. These products are sold at relatively high prices compared with the imported products for high quality of products and consumer preference for local products.

يقدر الانتاج المحلي من الجميد بحوالي 2,898 طن وينتج في العديد من المعامل والمنازل والجمعيات التي تنتج الجميد بطريقة تقليدية، والمنافسون الرئيسيون للجميد البلدي في محافظة الكرك هم: جمعية شفا الخير، وجمعية الكرك لتربية المواشي/مصنع أدر، وجمعية منشية أبو حمور ومصنع زحوم للجميد ومشتقات الألبان.

The local production of AlJamid is estimated about 2,898 tons and produced in many factories and houses that produce AlJamid in a traditional way. The main competitors of the AlJamid in AlKarak governorate are: Shafa Al-Khair Association, Al-Karak association for Livestock Breeding / Adar Factory, Manshiet Abu Hamour Association, and AlZahoum for AlJamid and dairy products.

لذا مايزال تصنيع الجميد يصنع على المستوى التقليدي باستثناء مصانع الكسيح وحمودة وطيبة واليوم الذين يقومون بتصنيع الجميد السائل بكميات تجارية. في حين تصنع الزبدة والسمنة من خلال المعامل الصغيرة والإنتاج الأسري.

Al Jamid is still manufactured at the traditional level, except for the factories of Alkaseeh, Hamouda, Taiba, and Al yaoum, which manufacture liquid Jamid in commercial quantities. While butter and margarine are made through small laboratories and family production.

اما مستوردي الجميد وهم شركة المواس، شركة ابو عودة إخوان، شركة خيرات الشام، محلات النهر الأبيض للجميد، شركة جوهرة الطفيلة، شركة منير عيس الملك، شركة محمد البغدادي، ومؤسسة محود ابراهيم. حيث بلغ عدد الرخص الممنوحة لاستيراد الجميد (31) رخصة بمتوسط حجم (100) طن للرخصة الواحدة.

Al-Jameed importers are Al-Mawas Company, Abu Odeh Brothers Company, Khairat Al-Sham Company, Al-Nahr Al-Abyad Gemayel Stores, Jawharat Al-Tafila Company, Mounir Issa Al-Malik Company, Mohammad Al-Baghdadi Company and Mahmoud Ibrahim Foundation. Where the number of licenses granted to import AlJamid is (31) with an average volume of (100) tons per license.

وجدير بالذكر أن الطلب المتزايد على الجميد الكركي يفوق ما يتم إنتاجه في المحافظة مما يزيد من الجدوى الاقتصادية لهذا المشروع لا سيما وأن السوق المحلي وحده يستوعب ما يزيد عن عشرة أضعاف ما يتم انتاجه حالياً، مما يجعل للمشروع مردوداً مادياً دائماً يفوق بكثير المردود الحالي.

It is worth mentioning that the increasing demand for Al-Jamaid Al-Karaki exceeds what is produced in the governorate, which increases the economical feasibility of this project, especially since the local market demand is more than ten times what is currently produced, which makes the project always cost-effective.

لذا لا يتوقع أن يواجه المشروع صعوبات في تحقيق المبيعات المتوقعة له على اعتبار ان الجميد من السلع الغذائية المرغوبة محليا وعلى مستوى الدول العربية لأهميتها الغذائية. ولا يزال الأردن بحاجة إلى مادة الجميد سواء للاستهلاك المحلي أو لتصنيع الجميد السائل. كما وأنه يتم استيراد الجميد لتغطية جزء من حاجة السوق المحلي، هذا ويمكن تسويق منتجات المشروع إلى الأسواق المحلية أو التصدير إلى الدول العربية والأوروبية.

Therefore, it is not expected that the project will face difficulties in achieving the expected sales, considering that Al Jamaid is one of the most desirable foodstuffs locally and at the level of Arabian

countries for its nutritional importance. Jordan is still in need of Al-Jamid, whether for local consumption or for manufacturing of liquid Jamid. Al Jamaid is also imported to cover part the local market demand. The products of the project can be marketed to local markets or exported to Arabian and European countries.

Market share

تم تحديد حصة المشروع من السوق والبالغة حوالي 100 طن سنويا من الجמיד من خلال استخدام 600 طن من حليب الأغنام في محافظة الكرك. وتم تقدير كمية انتاج محافظة الكرك من حليب الاغنام للتأكد من مدى توفر المواد الخام للتوريد للمصنع، حيث تبين بيانات مديرية الانتاج الحيواني في وزارة الزراعة ان عدد الأغنام في محافظة الكرك وألويتها قد بلغ 607 الف راس في نهاية عام 2015 وبذلك تشكل ما نسبته 14.3% من إجمالي عدد الأغنام في المملكة، تنتج سنويا ما مقداره 15,742 طن من حليب الاغنام (الضأن والماعز) وبذلك فان الكمية المطلوبة للمشروع من حليب الاغنام لا تشكل اكثر من 3.8% من انتاج محافظة الكرك من حليب الأغنام. وبذلك فانه لا خوف من عدم توفر الكمية المطلوبة من الحليب للمشروع من محافظة الكرك.

The market share of AlJamid through the use of 600 tons of sheep milk in AlKarak governorate was estimated at about 100 tons per year. The amount of production of sheep's milk was estimated to check the availability of raw materials needed for the plant. The number of sheep in AlKarak governorate and its sub-districts reached 607 thousand at the end of 2015, thus accounting for 14.3% of the total number of sheep in the Kingdom. Annually 15,742 tons of sheep's milk (sheep and goats) are being produced. Thus, the required amount for the project of sheep milk does not constitute more than 3.8% of the production of sheep milk in AlKarak governorate. Therefore there is no fear of the lack of the required amount of milk for the project from AlKarak governorate.

Table 8 Production of sheep milk in Karak. (Ton)

Karak/district	Number of sheep (head)	Number of goats (head)	Total of sheep (head)	Amount of sheep's milk (ton)	Amount of goat's milk (ton)	Amount of cattle's milk (ton)
Alqasaba	107,028	34,253	141,281	2,246	1,388	3,660
Alghour Aljnobi	24,783	52,261	77,045	520	2,118	1,996
Almazar Aljnobi	117,226	33,847	151,073	2,460	1,372	3,914
Alqutraneh	106,079	15,828	121,907	2,226	642	3,158
Alkaser	42,400	19,286	61,687	890	782	1,598
Faqou'	31,020	12,093	43,114	651	490	1,117
A'ee	6,081	5,443	11,524	128	221	299
Total governorate	434,618	173,013	607,631	9,122	7,012	15,742

Karak/district	Number of sheep (head)	Number of goats (head)	Total of sheep (head)	Amount of sheep's milk (ton)	Amount of goat's milk (ton)	Amount of cattle's milk (ton)
Total of kingdom	3,173,658	1,067,557	4,241,215	66,609	43,268	109,877

ويمكن من خلال ذلك تقدير حصة المشروع بناء على الطاقة الإنتاجية المتوقعة للمشروع والتي تصل إلى 3.45 % من الانتاج المحلي وكما يبين جدول (10) فان حصة المشروع من اجمالي الطلب الكلي على الجميد لا تتعدى ما نسبته 1.68 % من إجمالي الطلب. ويبين (جدول 11) حصة المشروع من السوق المحلي للجميد خلال الفترة (2018-2027)

It is possible to estimate the project's share based on the expected production capacity of the project which is 3.45% of the local production. As Table 10 shows, the project's share of total demand for Al-Jamid does not exceed 1.68% of the total demand. Table (11) shows the project's share of the local market for Aljamid during the period (2018-2027)

Table 9 Projected local consumption and production of AlJmeed

Jordan market size	2011	2012	2013	2014	2015	2016	2017
Imports (ton)	1,803	874	2,947	3,550	2,722	2,804	2,888
Exports (ton)	254	427	402	400	608	626	645
Re-exports (ton)	137	141	225	1	257	265	273
consumption (ton)	3,357	3,565	3,895	4,226	4,588	4,726	4,868
local production	1,945	3,259	1,575	1,078	2,731	2,813	2,898

Price analysis and pricing policy

يتم بيع الجميد الكركي من خلال المولات وتجار التجزئة، وإعتمادا على الاتصالات الشخصية بين المنتجين والمستهلكين ويتراوح سعر بيع واحد كغم من الجميد (10-18) دينار حسب المصدر والثقة التي يولها المستهلك بالمنتج. اما الجميد المستورد فتتراوح أسعاره ما بين (5-6) دينار الكيواغرم الواحد، ويتوفر في الأسواق حاليا الجميد السائل في عبوات ورقية مقواة سعة (1) لتر بسعر يتراوح ما بين (1.75- 2.25) من انتاج شركات الألبان المحلية.

Al Jamid Al-Karki is sold through malls and retailers, and depending on the personal contacts between producers and consumers. The price of selling one kilogram of Jamid ranges from (10-18) dinars depending on the source of the product. As for the imported AlJamid, the price ranges from (5-6) Kilograms per liter. Liquid Jamid is available in 1 liter paper packages at a price ranging from 1.75 to 2.25 for local dairy companies.

يتطلع المشروع الى انتاج جميد كركي بنوعية عالية جدا مطابقا للمواصفات الاردنية المعتمدة، بحيث يكون خالي من الشعر والأتربة مستخدما حليب النعاج والماعز الأصلي بدون إضافات او خلط مع حليب الأبقار او الجميد المستورد للصناعة، لإعطاء نكهة الجميد الكركي المميزة واستخدام نوعيات وجودة عالية من مدخلات الانتاج مع المحافظة على الشروط الصحية الجيدة للتصنيع والتخزين، لذا يقترح ان يكون سعر بيع ا كغم من الجميد بنوعية وجودة عالية لا

تقل عن (8) دينار للكيلوغرام الواحد من سعر الجملة للتجار ومحلات التجزئة وبيع للمستهلك بسعر (10-11) دينار للكيلوغرام الواحد.

The project aims to produce AlJamid AlKarak with a very high quality in accordance with the Jordanian specifications, so that it is free of hair and dust by using the milk of the ewes and goats without additives, mixing it with cow's milk or with imported Jamid. To give distinctive flavor and quality of the production inputs while maintaining good sanitary conditions for manufacturing and storage, it is proposed that the price of selling a kilogram of AlJamid with a high quality is (8) dinars of wholesale price for retailers and selling price to the consumer ranges from (10-11) dinars per kg.

Projected revenues

يمكن ان يتم انتاج الجميد على مدار العام في حال توفر حليب أغنام وحيث انه هناك فترة يكون فيها فائض وفترات يكون فيها شح من حليب الاغنام، يمكن ان ينتج المصنع مشتقات اخرى من حليب الابقار كاللبن واللبننة الطرية والجبنه البيضاء، حيث لا يتوقف المصنع عن الانتاج. ولكن لأغراض الدراسة تم التركيز على انتاج الجميد من حليب الأغنام وما يتبعه من منتجات جانبية تنتج جنبا الى جنب مع انتاج الجميد وهي الزبدة البلدية والسمن البلدي وبيّن جدول (9) حصة المشروع من السوق المحلي للجميد خلال الفترة (2018-2027) وبيّن جدول (10) المبيعات المتوقعة من المشروع. ولأغراض هذه الدراسة فقد اعتبر ان معدل سعر البيع (8) دينار/الكيلوغرام للجميد، (6) دينار لكل كيلوغرام من الزبدة و(5) دينار لكل كيلوغرام من السمن البلدي سعر باب المصنع بعد إجراء عمليات التعبئة والتغليف ويتوقع ان تزداد الاسعار بنسبة 2% سنوياً.

AlJamid can be produced throughout the year if sheep milk is available. Since there is a period of surplus and periods where there is a shortage of sheep milk, the factory can produce other dairy products such as yogurt, Labaneh and white cheese, as the plant does not stop producing. However, for the purposes of the study, emphasis was placed on the production of AlJamid from sheep milk and the other products, butter and margarine. Table (9) shows the project's share from the local market for AlJamid during the period (2018-2027). For the purposes of this study, it was considered that the average selling price is (8) JD / kg, (6) JD per kilogram of butter and (5) JD per kilogram of margarine and expected to increase by 2% annually.

Table 10 Project's share of AlJmeed for local market (2018-2027)

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
The expected population (Thous.)	9,984	10,204	10,428	10,658	10,892	11,132	11,377	11,627	11,883	12,144
Annual per capita consumption (kg/capita)	0.480	0.480	0.480	0.480	0.480	0.480	0.480	0.480	0.480	0.480
Domestic consumption of the product (tons)	4,792	4,898	5,006	5,116	5,228	5,343	5,461	5,581	5,704	5,829
demand of restaurants (ton)	500	511	522	534	545	557	570	582	595	608
Domestic demand of product (ton)	5,292	5,409	5,528	5,649	5,774	5,901	6,031	6,163	6,299	6,437
Expected production of the project (ton)	100	100	100	100	100	100	100	100	100	100
Market share of project to domestic demand (%)	1.89	1.85	1.81	1.77	1.73	1.69	1.66	1.62	1.59	1.55

Table 11 Production capacity of various dairy products (2018-2027)

Market share (ton)	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
First item-Al-jmeed	50	100	100	100	100	100	100	100	100	100
Second item-ghee	7.9	15.8	15.8	15.8	15.8	15.8	15.8	15.8	15.8	15.8
third item-butter	8.8	17.6	17.6	17.6	17.6	17.6	17.6	17.6	17.6	17.6
Total sales (tons)	67	133	133	133	133	133	133	133	133	133

Table 12 Sale prices and expected sales total during the period (2018-2027)

Project revenues	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total of sales (ton)	67	133	133	133	133	133	133	133	133	133

expected price for products (JD/ton)	8,000	8,160	8,323	8,490	8,659	8,833	9,009	9,189	9,373	9,561
Total sales (Thousand JD)	492	1,005	1,025	1,045	1,066	1,087	1,109	1,131	1,154	1,177

3. Technical analysis

ان الأساليب المستخدمة في صناعة الألبان القائمة على حليب الأغنام لا زالت تقليدية ولم يطرأ عليها أي تطور يذكر منذ عشرات السنين، وهي تقليدياً، يتم تخفيف اللبن الخائر المنتج وخض. ثم يتم تركيز مخيض الحليب الناتج (الشنيينة) باستخدام الشاش للحصول على منتج ذي قوام ثخين جداً ويضاف الملح بطرق وكميات مختلفة خلال مرحلة التصفية والتركيز. يتم تشكيل المنتج الناتج على شكل كرات مستديرة ثم يترك لبضعة أيام ليجف.

The methods used in the dairy industry based on sheep's milk are still traditional and have not undergone any development for decades. Traditionally, milk is diluted. The resulting milk is then concentrated using gauze for a product with a very thick texture then salt is added in different quantities during the filtration and concentration phase. The resulting product is formed as a rounded balls and then left for a few days to dry.

وهناك طرق معروفة لصناعه كل منتج من منتجات الألبان ونظرا لتشعب هذه الصناعة فلن نخوض في تفصيل العملية الإنتاجية لكل منتج، ومن المتعارف عليه أن أي مصنع ألبان يجب أن يشتمل على وحدة لاستلام الحليب وتنظيفه، ثم المخازن المبردة وآلات التجهيز والتسخين والتعقيم والمجانسة، ثم المنشآت وتشمل الغرف الحاضنة والمبردة ومستودعات لمواد التغليف والتعبئة. ويمكن وصف الوحدات الإنتاجية لمصنع الألبان كالتالي:

There are ways to manufacture each dairy product. Due to the divergency of this industry, we will not elaborate on the production process of each product. It is common that any dairy factory should include a unit for receiving and cleaning milk, refrigerated stores, processing machines, heaters, sterilization and homogenization. Facilities include incubator rooms, refrigerated and warehouses for packaging materials. Dairy production units can be described as follows:

- قسم استلام الحليب ووزنة وفحصه مع وحدة تنظيف وتشمل على ميزان، ومصافي للحليب ومضخات ومبادل حراري
- قسم البسترة والمعاملات الحرارية
- قسم الأجبان البلدية، وتشمل على أحواض تجبين ومكابس
- قسم تصنيع الجميد، وتشمل على أحواض، ومكابس وأماكن تقطيع وتشكيل وتجفيف
- مستودعات الإنضاج
- قسم معدات الطاقة (البخار، الكهرباء، التكييف والتبريد) وكذلك وحدات التنظيف.

- Receiving, weighting and inspecting of milk departmen: includes a balance, milk filters, pumps and heat exchanger.
- Pasteurization and thermal treatment department.
- Cheese section, including basin and presses.
- Al-Jamid Manufacturing Department, which includes basins, presses, cutting, forming and drying areas.
- Maturation warehouses.

- Power Equipment Department (steam, electricity, air conditioner and refrigerators) as well as cleaning units.

قدرت الطاقة الانتاجية القصوى للمعمل ب 33 طن من الزبدة والسمنة و 100 طن من الجميد سنويا وذلك بوردية عمل واحدة و300 يوم عمل في السنة. هذا وقد تم افتراض أن المشروع سيبدأ تشغيله بنسبة طاقة مستغلة 25% للسنة الأولى على أن تزداد كميات الإنتاج ليصل الى كامل الطاقة الانتاجية في السنة الثانية.

The maximum capacity of the plant was estimated at 33 tons of butter and ghee and 100 tons of Jamid per year with one working shift and 300 working days per year. It is assumed that the project will start operating at a 25% capacity for the first year, with increasing to full capacity in the second year.

The required raw materials for the project

يلزم لانتاج 100 طن من الجميد حوالي 600 طن من حليب الإغنام بسعر شراء 0.8 دينار للكيلوغرام، علما بأن اسعار حليب الأغنام تبدأ من (1) دينار في بداية الموسم (كانون ثاني – شباط) وتنخفض الى (0.6) دينار في ذروة موسم الحلابة (آذار، نيسان و أيار) وتم الافتراض (0.8) دينار كمتوسط سعري للحليب الطازج من المربين في مواقع انتاجهم، اي ان المصنع سوف يتحمل كلفة النقل والتجميع. وتتمثل كذلك المواد الأولية الرئيسة بالإضافة الى حليب الأغنام على ملح الطعام، عبوات زجاجية، عبوات كرتون ونكهات اعشاب (تحويجة) ، ويحتاج كل كغم مصنع من المنتجات المذكورة في الدراسة إلى مكافئ من الحليب كما هو مبين في الجدول (12) ويبين جدول (13) كلفة المواد الأولية للصناعة والكمية السنوية لانتاج 100 طن من الجميد وبقية مشتقات الألبان.

To produce 100 tons of Jamid, about 600 tons of sheep milk at a purchase price of 0.8 dinars per kilogram is required. The price of sheep's milk starts from 1JD at the beginning of the season and falls to 0.6JD at the peak of the milking season (March, April and May), and a (0.8) dinars was assumed as the average price of fresh milk from the breeders at their sites, thus, the plant will bear the cost of transportation and collection. As well as the main raw materials of sheep's milk, and salt, glass containers, cardboard containers and herbal flavors. Each kilogram of the mentioned products needs an equivalent quantity of milk as shown in Table 13. Table (14) shows the cost of raw materials and the annual quantity of 100 tons of Jamid and other dairy products.

Table 13 the amount of milk to produce 1 kg of the product

Product	Milk cows	Sheep milk
Yogurt	1	1
Labneh	4	2
Cheese	6.5	4
Jamid	12	6
Ghee	25	19
Butter	22	17

المصدر: مصنع الألبان التجريبي في الجامعة الأردنية

Table 14 Prices, quantities and the cost of raw materials required annually

Raw materials	Price (JD/kg)	Amount (kg)	Value (JD)
Fresh sheep milk	0.80	600,000	480,000
Salt	0.50	5,000	2,500
Herbal Flavors	1.50	3,000	4,500
Carton	0.80	8,000	6,400
Glass containers	0.20	26,520	5,304
Plastic bags	0.20	50,000	10,000
Total			508,704

The required human resources

يحتاج هذا المشروع الى مهندس زراعي تخصص تصنيع غذائي بالإضافة الى فني تصنيع البان او فني تصنيع زراعي وكذلك مندوب يتولى عملية توريد الألبان الى المصنع وتسويق المنتج الى الأسواق المستهدفة وسائقين لجلب الحليب وعمال تصنيع في المصنع. ويبين جدول (14) الاحتياجات من العمالة وأجورها.

This project needs an agricultural engineer pecialized in food processing as well as a dairy manufacturing technician or an agricultural processing technician as well as an officer who handles the supply of milk to the factory and marketing the product to the targeted markets and drivers to bring milk and manufacturing workers in the factory. Table (15) shows the required labor for the project.

Table 15 the required labor for the project

Job title	No.
Indirect staff	
Project Manager	1
Agricultural engineer	1
Food processing technician	2
Marketing and supply officer	1
Security	1
Total Non-Direct staff	6
Maintenance technician	1
Laboratory technician	1
Technical production supervisor	2
Workers and drivers	6
Cleaners	2
Total of direct staff	12
Grand Total	18

تشمل التكاليف الإدارية رواتب المدير التنفيذي للمشروع، موظفي المبيعات، وفي التصنيع الغذائي والحراسة لـ... لتصل التكلفة المرتبطة بالتكاليف الإدارية إلى (52,000) دينار في السنة، وتزداد رواتب العمالة المباشرة نتيجة تراكم الخبرة لتصل إلى 2 % سنوياً، وعلاوة على هذا يتم إضافة التأمين الصحي للعمال ومساهمة الضمان الاجتماعي وبدل التنقل بحيث تم احتساب الرواتب السنوية لجميع العاملين بمقدار 16 شهر بدل من 12 شهراً لتغطية هذه التكاليف وذلك لتقدير إجمالي أجور العمالة السنوية بما فيها أجور العمالة غير المباشرة كما هو مبين في الجدول (15)

The administrative costs include the salaries of the project's executive director, sales staff, food processing technician and guards, etc. The cost associated with the administrative costs is (52,000) JD per year. The direct labor salaries increase as a result of accumulated experience to reach 2% per year. In addition to the social security and transportation so that the annual salaries of all employees were calculated by 16 months instead of 12 months to cover these costs in order to estimate the total annual labor wages including indirect labor wages.

Job description:

Agricultural Engineer/food processing: Preparation of the production plan according to the type of food products to be manufactured, their quantity and schedule, the number of workers and the machines used, and the development of appropriate technical solutions to the problems related to the production process or the quality of the product. Determination of the raw materials involved in product composition, in terms of type and percentage of use and compliance with them, follow up the process of receiving milk and additives, and weight, and check their validity, and matching specifications, and keeping them in refrigerators. Supervision of the status of milk in dairy products and equipment, such as: pasteurizer, fat sorting machine, cream extract, butter, chickpeas, cheese presses, cheese forming machine, set and operated. Supervising the manufacture of different types of milk and dairy products, such as: yoghurt, Libneh, butter, cream, white cheese, liquid Jameed and solid Jameed. Evaluate the performance of staff and labors, identify their training needs, train them and develop their skills. Mobilization of business models. Applying occupational safety and health procedures and instructions.

Food Processing Technician: Processing the required quantities of milk for each production line, and matching specifications. Processing of machines and equipment Pasteurization unit, such as: Pasteurizer and automatic separator, adjust the temperature, duration, pressure, and operation. Monitor device controls, such as: pasteurization device. Cleaning and sterilizing equipment, tools and pasteurization unit. Receiving the raw materials necessary for the manufacture of the Jamid and butter, flavors, weight, check their validity, and meet their specifications. Adjust the cooker flame, add the butter gradually and

additional materials, such as: Borgul and flavors on the butter during boiling, stirring, and cooling of the produced fat. Packaged with appropriate packaging according to the required weights, and storage in appropriate storage conditions, such as: temperature and humidity and light. Clean the cooker and tools used. Applying occupational safety and health procedures and instructions. Put the milk in the sorting machine, adjust the excretion, operate it, monitor the sorting process, and transfer the sorted cream to the cooling system. Development of procedures and methods of work and production using modern technologies. Development of quality systems and standards. Prepare technical reports on food processing operations and implement procedures and ensure occupational and public safety and health requirements.

Marketing, Supply and Accounting Representative: Development of plans for the production, sale and marketing of products, the development of regulatory procedures and instructions for production control, supervision of the organization of daily work and the distribution of staff and labors. The annual budget for the supply of milk to the intended production, the selection of improved and good qualities, control of records of production and sale, revenue and expenditure. Monitoring and developing quality systems and standards. Secure setting for the production, sale and marketing of products, plans, requirements, and to develop regulatory procedures and instructions to control the production, preparation for receipt of products and inspection and matching quantities, types and specifications with the purchase order, and check their suitability for human consumption, and special packaging process of receipt and delivery models, and follow-up loading and unloading and receiving process, and the classification of special packaging product and arrange them according to type, install selling prices, explain the advantages and specifications of the different product types, and help customers to choose their needs, and to respond to their queries, and securing their applications and follow-up, and the weight of materials, and packaging according to customers' request, sell product, customer accounting, billing, inventory the contents of the stores. Identifying the needs of the company for agricultural production requirements, and preparing lists of shortages, insurance and storage. Regulation of sales records and financial records, mobilization of business models and application of occupational and public safety and health procedures and instructions.

Laboratory Supervisor: Supervise the receipt of fresh milk containing high-fat, and take a sample to examine the proportion of fat. Monitoring chemical and biological pollution, taking a sample of each product type, and checking it to ensure that the product meets the required specifications such as pH and salinity. Follow up the supervision of the cleaning of dairy products and equipment, and determine the appropriate temperature and humidity to store raw materials and food products and the validity period,

and match the results of laboratory tests of the samples with quality specifications, filling the work models. Application procedures and instructions of the occupational safety and health.

Production Supervisor: works under the supervision of the agricultural engineer and supplies raw materials and conforms its specifications, preparing machines and equipment for the manufacture of dairy products and their derivatives, processing dairy products, inspecting the conformity of the product to the required specifications, cleaning the equipment, tools, packaging, and store them until marketing. Processing of packaging tools and accessories of packaging. Placing the packagings on the production line, sorting the product, and classifying it according to the required specifications. Fill the product in the custom packaging, and pack it according to instructions. Collecting and packaging of containers, and store them in warehouses until marketing.

Dairy factory worker: Works under the supervision of a dairy manufacturer and receives the raw materials needed to manufacture the Jameed, and preparing the tools and equipment for the manufacturing process, such as: mixing and shaking machines and operate them, and the manufacture and drying and storage of Jameed and cleaning of used tools and equipment in the production, and the application of occupational safety and health procedures and general. Receiving the raw materials necessary for the manufacture such as: milk and yoghurt, and weight, and check the validity period, and meet the specifications. Place the yoghurt in the mixture or mix. The addition of salt to a certain percentage, the formation of the resulting pulp in the form of tablets, placing it on a table for drying, and the storage of dry products in appropriate storage conditions such as temperature and humidity. Cleaning of tools and equipment, such as mixer. Applying occupational safety and health procedures and instructions.

Maintenance and Repair Technician: Works independently or under the supervision of the agricultural engineer and maintains machinery and equipment, valves and supplies, ensuring water pumping, maintain and repair machinery and equipment, follow up the procedures of installation and maintenance of machines, preparation of appropriate operating programs; preparation for the machines and equipment cleaning and sterilization; and determine the materials allowed to be used in Cleaning and sterilization operations. Training workers on how to operate the machines and turn them off and maintenance of warehouses and storage rooms of production supplies and ready-made goods. Knowledge of the correct ways to clean machines, define the specification and requirements of each machine separately. Inspect the materials, equipment and tools needed for the cleaning process, such as brushes, cloths, brooms, etc. Inspect the machines for the cleaning process in coordination with the operators of the machines, cleaning the machines according to the instructions of the manufacturers, collecting the waste of the cleaning process in the places allocated to it, and discharging the liquid waste

according to the instructions. Maintain the cleanliness of the work site, and apply the procedures and instructions of occupational and public safety and health.

Site analysis

بناء على جميع هذه المعطيات فإنه يقترح أن مصنع للجميد الحجر والسائل في محافظة الكرك بحيث يتم اختيار الموقع الملائم للإنتاج بناء على توفر خدمات النية التحتية من ماء وكهرباء وتوفر المواد الأولية والنظام الإنتاجي السائل، فعلى سبيل المثال يمكن إقامة مصنع لتصنيع الجميد من حليب الأغنام في الكرك القصبة أو المزار الجنوبي أو القطرانة نتيجة تواجد اعداد كبيرة من الاغنام وتوفر المادة الخام ولكن النقطة الحساسة في هذه المشاريع هو توفر وسائل النقل المبرد لنقل الحليب من المربين إلى المصنع، ويمكن عمل ذلك باستخدام ثلاجات صغيرة متنقلة لنقل الحليب إلى موقع المشروع. وخلال فترة الربيع الممتدة من شحراكانون ثاني إلى نيسان وتموز يمكن أن يعتمد المصنع على تصنيع منتجات حليب الأغنام ، وخارج هذا الموسم يمكن أن يصنع حليب الأبقار مع حليب البودرة.

Based on all these data, it is proposed that the Jamid plant will be established in AlKarak governorate and the selection of the suitable location for production should be based on the availability of the infrastructure services of water, electricity, the availability of raw materials and liquid production system. For example it is possible to establish a plant for manufacturing Jamid in the region of Al Qasabah or the AlMazar Al Janoubi or Qatranh due to the presence of large numbers of sheep and the availability of raw material, but the critical point in these projects is the availability of refrigerated transportation means to transport milk from the breeders to the plant, this can be done by using portable mini-refrigerators for the transportation of the milk to the project site. During the spring period from January to April, the factory can rely on manufacturing products of sheep milk, and in other seasons, cow's milk with powdered milk can be produced.

Table 16 Proportion and the numbers of cattle in AlKarak governorate

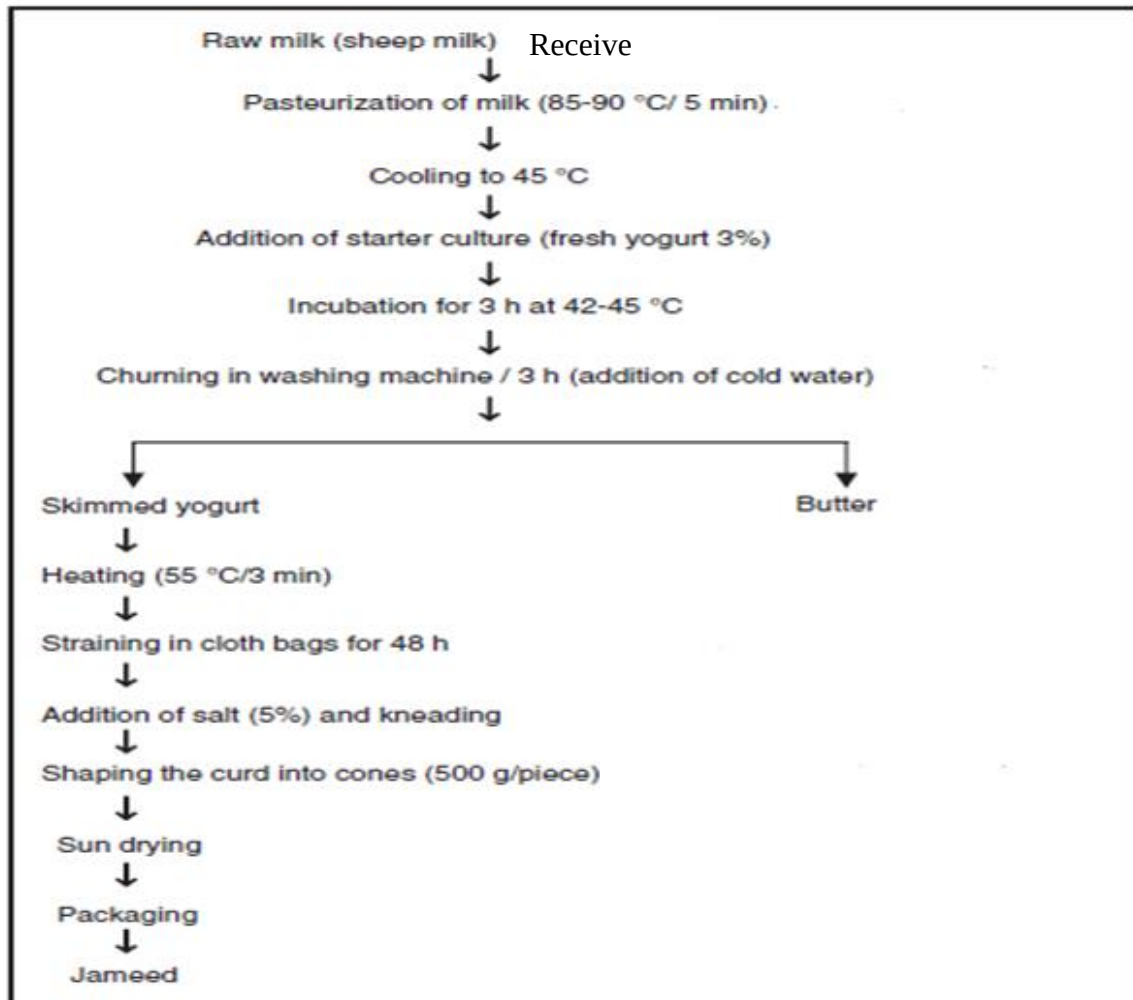
Karak/District	Numbers of sheep			Percentage		
	Sheep	Goat	Total cattle	Sheep	Goat	Cattle
Qasaba	107,028	34,253	141,281	3.4%	3.2%	3.3%
Al-Aghwar jnobi	24,783	52,261	77,045	0.8%	4.9%	1.8%
Al-Mazar jnobi	117,226	33,847	151,073	3.7%	3.2%	3.6%
Al-Qutranah	106,079	15,828	121,907	3.3%	1.5%	2.9%
Al-Qassr	42,400	19,286	61,687	1.3%	1.8%	1.5%
Faqou'	31,020	12,093	43,114	1.0%	1.1%	1.0%
A'ee	6,081	5,443	11,524	0.2%	0.5%	0.3%
Total	434,618	173,013	607,631	13.7%	16.2%	14.3%
Total of Kingdom	3,173,658	1,067,557	4,241,215	100.00%	100.00%	100.00%

Description of the production process and manufacturing steps

يعتبر الجميد والزبدة والسمن البلدي من المنتجات الشعبية والمعروفة في الأردن، حيث تم تصنيعه واستهلاكه منذ قرون عديدة. وعملت إيكاردا مع المركز الوطني للبحث والإرشاد لإدخال بعض التغييرات في عملية تصنيع الجميد التقليدية حيث أن الطريقة المعدلة تحسن من جودة المنتج وتقلل من استهلاك المياه والطاقة اللازمة في التصنيع. اعتمدت هذه التغييرات على إدخال بعض التقنيات البسيطة والتي يستطيع المصنع تطبيقها والقيام بها. وتشمل عملية الانتاج من البداية بإستلام الحليب، ووزنه وتصفيته من الشوائب، ثم إجراء عملية البسترة بغليه على درجة حرارة (85-90 مئوية) لمدة خمس دقائق ثم تبريده وبعدها إضافة البادئ (3% لبن طازج) ثم فترة تحضين لمدة (3-5) ساعات على درجة حرارة (42-45) مئوية، وبعدها إجراء عملية الخض (كحوض الغسالة) لمدة 3 ساعات (وإضافة الثلج أو الماء البارد لتنفصل الزبدة عن شرش أو مصل الحليب للحصول على حليب خالي الدسم (حليب منزوع الدسم) ثم تأتي عملية تسخينه على درجة حرارة (55) مئوية لمدة 3 دقائق، ووضعه في أكياس لمدة 48 ساعة أو جهاز لنزع الماء، وبعدها يضاف الملح بنسبة (3%)، وبعدها تشكيله على شكل أقراص حجرية وتجفيفه وتغليفه في أكياس بلاستيكية مفرغة من الهواء تمهيدا لتخزينه وتسويقه لاحقاً (شكل 1).

Jamid, butter and ghee are popular products known in Jordan, and had been manufactured and consumed for decades. ICARDA has worked with the National Research and Extension Center to introduce some changes in the process of manufacturing AlJamdi, as the modified method improves the quality of the product and reduces the water consumption and energy needed in manufacturing. These changes have been based on the introduction of some simple techniques that the manufacturer can apply and carry out. The process of production from the beginning of receiving milk, and weighting and clear it from impurities, and then Pasteurization process by boiling at a temperature of (85-90) for five minutes and then cooled and then add 3% fresh milk, then the incubation period (3-5) at a temperature of (42-45) and then the mixing process for 3 hours and ice or cold water is added to separate the butter from the whey to get skimmed milk and then heating it at a temperature of (55) for 3 minutes, and putting it in bags for 48 hours or a device for dehydration, and after 3% of salt is added and then form it in the form of balls and dry it and wrap it in plastic bags emptied from the air for storage and marketing (Fig. 1).

Figure 1 AlJamdi production process by the traditional way.



ويتكون مصنع الألبان من عدة أقسام أهمها:

Dairy factory is consists of several departments:

1. قسم تسلّم الحليب الخام وتخزينه مبرداً: يتم في هذا القسم فحص حموضة الحليب التي يحدد بموجبها قبول الحليب أو رفضه، ثم وزن الحليب وتبريده وتخزينه في خزانات كبيرة الحجم إلى حين التصنيع.

Department of receiving and strage of raw milk cooled: In this department, the milk acidity is tested which determines the acceptance or rejection of the milk, then weighting and cooling the milk and stored in large tanks until manufacturing.

2. قسم الإنتاج: وهو أكبر أقسام مصنع الألبان، إذ يتألف من عدة صالات للإنتاج حسب عدد خطوط الإنتاج التي في المصنع.

Production Department: It is the largest department in the dairy factory, consisting of several production lines.

3. قسم التخزين يتكون من قسمين: الأول؛ ويتألف من عدة غرف كبيرة مبردة لتخزين المنتجات اللبنية الجاهزة بعد تصنيعها إلى حين تسويقها. والثاني؛ يتألف من عدة مستودعات غير مبردة لتخزين المواد المساعدة في الإنتاج، مثل مواد التعبئة والتغليف من زجاجات الحليب المعقم، أو العبوات اللدائية أو المعدنية، وغيرها من مستلزمات الإنتاج. القسم الخاص بالمخبر: يجهز بجميع الأجهزة والأدوات والمواد الضرورية لفحص الحليب الخام وتقييمه، وكذلك المنتجات الجاهزة وتحديد نوعيتها ومدى مطابقتها للمواصفات القياسية بكل منتج.

Storage Department consists of two parts

The first consists of several large refrigerated rooms to store ready-made dairy products after they are manufactured until they are marketed. The second part consists of several non-refrigerated warehouses for storing auxiliary materials in production, such as packaging materials from sterile milk bottles, plastic or metal packaging, and other production requirements.

Laboratories: It equipped with all the necessary equipment, tools and materials for checking and evaluating raw milk, as well as finished products, their quality and their conformity with the standard specifications of each product.

4. قسم الإدارة والتسويق والخدمات المختلفة الأخرى: ويضم مكاتب الإدارة والمرافق العامة من مطاعم وحمامات وصالة للبيع.

Management and Marketing Department: includes the administrative offices, public facilities, restaurants and bathrooms.

5. قسم إعداد المياه المستخدمة في معامل الألبان

ويقسم إلى ثلاثة أقسام، هي:

أ. قسم معالجة المياه لكي تصبح صالحة للاستخدام في المصانع؛ وذلك باتباع سلسلة من العمليات الآلية والكيميائية بالترتيب الآتي: - الترسيب والترشيح للتخلص من المواد العضوية العالقة بالمياه. - إزالة عسر المياه الناتج من وجود كربونات الكالسيوم؛ والذي يسبب وجودها مشكلات عديدة، وأهمها ترسيب طبقة كلسية على سطح ألواح التسخين مما يقلل الناقلية الحرارية لهذه الألواح، كما أن العسر يسبب زيادة كمية مواد الغسيل الضرورية. - تعقيم المياه بالكلور للقضاء على الأحياء الدقيقة الممرضة.

ب. قسم التزويد بالماء الساخن والبخار الضروريين لعمليات التصنيع والغسيل، وذلك عبر دائرة كاملة للتسخين، والتي تتضمن المرجل وملحقاته الضرورية.

ج. قسم التزويد بالماء البارد الضروري لتبريد منتجات الألبان بعد تصنيعها، أو لتبريد الحليب في دائرة كاملة للتبريد بعد بسترتها أو تعميمه.

Water treatment Department: In order to become usable in the plant, by following a series of automated and chemical processes in the following order:

1. Sedimentation and filtration to dispose organic matter related to water.
2. Remove the water shortage resulting from the presence of calcium carbonate, which causes many problems, the most important of which is the deposition of calcareous layer on the surface of the heating panels, which reduces thermal conductivity of these panels, and the water shortage causes the increase in the amount of washing.
3. Sterilization of chlorinated water to eliminate pathogenic microorganisms.

The supply of hot water and steam necessary for manufacturing and washing operations, through a complete heating circuit, which includes the boiler and its necessary accessories.

Provision of cold water, necessary for dairy products cooling after being manufactured, or to cool the milk in a full cooling cycle after sterilization.

6. خطوط الإنتاج وتتضمن أجهزة ومعدات مختلفة الوظائف ويمكن تصنيفها وفق الآتي:

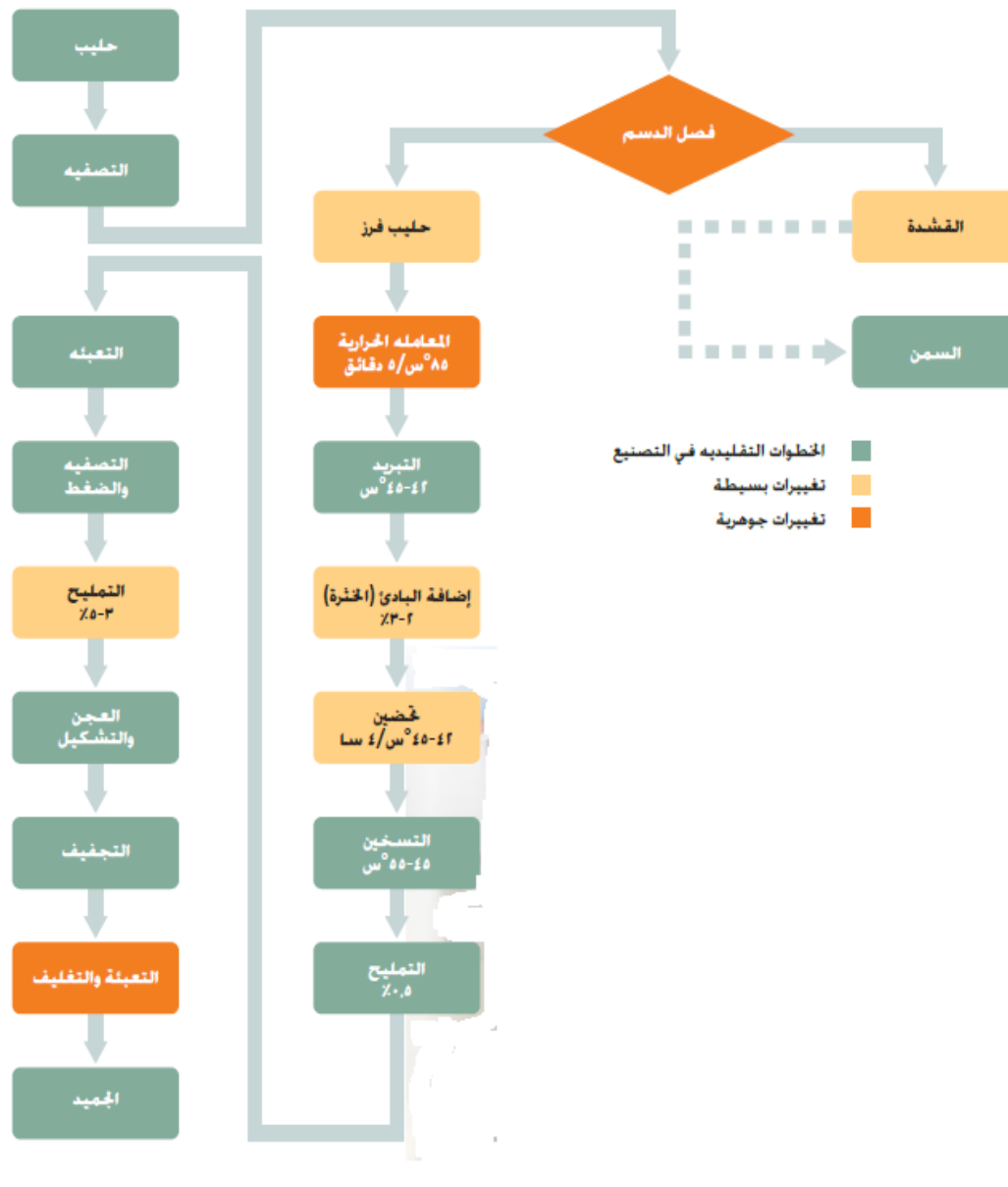
- أجهزة ومعدات عامة تستخدم في خطوط إنتاجية مختلفة، وتشمل: أحواض تخزين الحليب الخام، ومبادلات حرارية صفائحية وأنبوبية لتسخين الحليب وتبريده، وفراغات منقية وأخرى للدهن، وأنابيب وزوايا معدنية لربط الأجهزة مع بعضها بعضاً، ومضخات لنقل الحليب، وصمامات للتوازن، ودائرة التبريد والتسخين للماء.
- أجهزة ومعدات خاصة بكل منتج، وهي متنوعة بحسب المنتج، مثل الخضاض لصناعة الزبدة والجميد، كما يزود كل خط بآلة تعبئة وتغليف ولاسيما بالمنتج المصنّع. وكل هذه المعدات والأجهزة يجب أن تصنع من معدن غير قابل للصدأ ستانلس ستيل. والشكل رقم (2) خطوط الإنتاج في مصانع الألبان.
- ويبدو أنه هناك مجال واسع لإدخال تحديثات جذرية على هذه الأساليب، حيث قام المركز الدولي للبحوث الزراعية في المناطق الجافة (ICARDA) باختبار طريقة التصنيع المحسنة بنجاح في حزيران ٢٠١٢ مع منتجين من منطقة الكرك، الأردن. أدخلت الطريقة الجديدة التغييرات التالية، والتي تعتبر ثلاثة منها تغييرات جوهرية مشار إليها باللون البرتقالي.

6. **Production lines:** They include different devices and equipment of different functions and can be classified as follows:

General equipment and devices used in different production lines, including: raw milk storage basins, plate and tubular heat exchangers for milk heating and cooling and purificators and grease separators, metal pipes for connecting devices together, milk transfer pumps, balance valves and water cooling and heating circuit. Specific equipment to each product, such as the pulp for the butter and Jamid manufacturing. Each line is also supplied with a filling and packaging machine. All this equipment must be made of stainless steel. And Figure (2) shows the production lines in dairy factories.

It seems that there is a wide scope for radical updates for these methods. Whereas, the International Center for agricultural research in the dry area (ICARDA) had successfully tested the improved manufacturing method in June 2012 with producers from AlKarak governorate. The new method introduces the following changes, three of which are significant changes and are marked in orange.

Figure 3 Steps to manufacture the process in an improved manner



The analysis of technical requirements

Land and construction

يحتاج المشروع إلى مساحة إجمالية تقدر ب (2500) مترا مربعا يخصص 460 مترا مربعا للمبنى وصالة التصنيع ، والمساحة الباقية تخصص للخدمات والمرافق والممرات. ويقدر سعر المتر المربع بحوالي (20) دينار للمتر. وبذلك تكون كلفة الأرض بمقدار (50,000) دينار.

The project needs a total area of (2500) square meters, which allocates 460 square meters for the building and production hanger, and the remaining space for services, facilities and corridors. The estimated price at (20) dinars per square meter. Thus, the cost of the land is JD (50,000).

يحتاج المشروع إلى الأبنية والمرافق التالية وهي مكتب إدارة، صالة انتاج، مخازن مبردة، غرف تخزين ومختبر. وتشمل التكلفة الأعمال الإنشائية المطلوبة للمشروع من تصميم المبنى والمخططات اللازمة للمشروع وأعمال الحفر والتعبيد والساحات والكراجات. والمبينة في الجدول (17) Error! Reference source not found. (found). كما يلي:

The project needs the following buildings and facilities: management office, production hanger, refrigerated warehouses, incubator rooms and laboratory. The cost includes construction work required for the project from the design of the building and the necessary blueprints for the project, digging, paving, and garages. Shown in table 18 as follows:

Table 17 the required construction work

Construction work	Area (m ²)
Administrative offices	50
Lab	30
Production hanger	300
Incubated room	30
Storage warehouse	50
Sattling	2,000

Machinery and Equipment

The following table shows the required machinery and equipment needed for the project:

Table 18 Machinery and equipments

Machinery and equipment	Quantity
Steam boiler	1
Pasteurization device	1
Milk containers	100
Tables and clips made of stainless steel	4
Stainless steel tanks capacity of 1 m ³	2
tables with racks	10
Fat separating machine	2
cooling tanks of Stainless steel, capacity 2 tons	2
مبرد صفائحي	1
Pipes	1
Generator	1
Special equipments for the cooling room	1
Others (water tanks, pumps, tools. etc.)	1
Oven	1
Vacuum packaging machine	1
Lab tools	1
Other tools	1

Laboratory equipment includes sensitive balances, water bath, centrifuge, lactometer, PH meter, glassware, chemicals and other necessary tools to check for bacterial contamination and quality of production.

Furniture

The project needs simple furniture for management and labor estimated at about 3000 JD, which consists of tables, offices, lockers, laboratory furniture and a restroom.

Vehicles and modes of transportation

The following table shows a list of vehicles and modes of transportation required for the project

Table 19 Vehicles and modes of transportation

Vehicles and modes of transportation	Quantity
Refrigerated vehicles	2
Medium vehicle (pick-up)	1

Essential services

تتوفر خدمات الكهرباء والمياه والمواصلات في معظم مناطق المملكة، كما افترض توفرها في موقع المشروع. هذا وقدرت احتياجات المشروع من الخدمات الضرورية لتشغيل الآلات والمعدات (كهرباء ومياه، ديزل ومحروقات وبخار) بحوالي 2% من إجمالي الإيرادات السنوية كما هي مبينة في الجدول (20)

Electricity, water and transportation services are available in most parts of the Kingdom and are assumed to be available at the project site. The needs of the project (electricity, water, diesel, fuel and steam) are estimated to be about 2% of the total annual revenues as shown in Table (20)

Table 20 Needs of support services for project

Expenses assumptions	
Utilities expenses	2.00%
Maintenance expenses	0.10%
Consumables expenses	0.50%
Insurance expenses	0.05%
Marketing expenses	1.00%

Legal requirements

ان نشاط تصنيع الألبان يحتاج الى تراخيص ورقابة دائمة وصارمة كون المشروع يسعى الى انتاج سلعة غذائية محضرة للأسواق المحلية والتصدير الى الاسواق العربية والأجنبية، لذا يجب قبل البدء في الاستثمار استيفاء عدد من الشروط القانونية والإدارية. وتشمل هذه تسجيل منشأة قانونية، والحصول على التراخيص والموافقات اللازمة وغيرها. وتسليم المستندات المطلوبة ودفع التكاليف الإدارية الأولية المرتبطة بالاستثمار.

The dairy industry needs constant and strict licenses and control. The project seeks to produce a prepared foodstuff for local markets and exports to Arabian and foreign markets. Therefore, before

starting the investment, a number of legal and administrative conditions must be met. These include the registration of a legal establishment, the obtaining of necessary licenses and approvals, etc. Providing the required documents and payment of the initial administrative costs associated with the investment.

1. التسجيل القانوني سواء كانت مؤسسة فردية او شركة (تضامن، توصية بسيطة، مساهمة خاصة، شركة ذات مسؤولية محدودة، الخ)
2. تسجيل الاسم التجاري، للشركة التي تطلب تسجيل الاسم التجاري
3. التسجيل لدى غرفة الصناعة / التجارة. يجب تسجيل كل منشأة تعمل في مجال النشاط الصناعي أو التجاري في غرفة الصناعة / التجارة
1. Legal registration whether an individual institution or a company (solidarity, simple recommendation, private contribution, limited liability company, etc.).
2. Registration of the trade name of the company.
3. Registration at the Chamber of Industry & Trade. Every establishment in an industrial or commercial activity must be registered in the Chamber of Industry & Trade
4. الترخيص المهني - شرط أساسي للمنشآت القانونية لتكون قادرة على القيام بأنشطة الأعمال.
5. تسجيل في المؤسسة العامة للضمان الاجتماعي - تلتزم كل منشأة قانونية بموجب أحكام القانون الأردني للتسجيل في الضمان الاجتماعي
6. يلزم الحصول على موافقة مؤسسة الغذاء والدواء، لتداول المنتج بعد دراسة مكوناته وبطاقة بيانه واستعماله، حيث تتولى مؤسسة الغذاء والدواء التأكد من توفير الاشتراطات الصحية لضمان ثبوت صلاحية المنتج للاستهلاك البشري
4. Professional licensing - a prerequisite for legal establishments to be able to carry out business activities.
5. Registration at the General Foundation for Social Security - Each legal establishment is bounded by the provisions of Jordanian law to register in social security
6. The approval of the Food and Drug Administration is required for the circulation of the product after studying its components and declaration card and its use. The Food and Drug Administration shall ensure the provision of health requirements to ensure the validity of the product for human consumption

4. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 11.97% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 0 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 21 General Assumptions

All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 22 Currency Exchange Rates

American Dollar	0.705
Euro	0.754

Table 23 Annual Growth Rates Assumptions

Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	2.00%
Annual Expenses Increase Rate	2.00%
Annual Increase Rate in Employees Salaries	3.0%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Risk Free Rate of Return (1)	6.50%
Return to Debt Maturity (2)	7.50%
Market Risk Premium (3)	9.93%
Income Tax Rate	0.00%
Beta (4)	1
	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodaran Beta By Sector, , pages.stern.nyu.edu

Table 24 Income Tax Assumptions

Average Income Tax in Jordan(1)	5%
Income Tax Deduction(2)	100%
Income Tax after Deduction	0%

Compulsary Reserve Percentage(3) 10%

Annual Monthly Salaries have been calculated after multiplying monthly salaries 16

Table 25 Weighted Average Cost of Capital

Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%7.50
Market Risk Premuim	%9.93
Income Tax Rate	%0
Bets	1.00
Equity	201,186
Loans Value	201,186
Cost of Borrowing before Tax	%7.50
Cost of Borrowing After Tax	%7.50
Loans Value	201,186
Loans Percentage	%50
Cost of Equity	%16.4
Equity Value	201,186
Equety Percentage	%50
Project Value	402,373
Proportion of the project	%100
Weighted Average Cost of Capital	%11.97

Capital Expenditures

The estimated cost of the project is 402,373 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 26 Land Capital Expenditure

Required Land	20	2,500	50,000
Cost			50,000

Table 27 Construction Work Capital Expenditure

Administrative offices	150	50	7,500
Lab	150	30	4,500
Production hanger	150	250	37,500
Incubated room	150	30	4,500
Storage warehouse	150	50	7,500
Sattling	2	2,000	4,000
Other works	-	-	6,150
Total			71,650

Table 28 Machinery and Equipment Capital Expenditure

Pasteurization device	1	10000	10,000
Milk containers	1	15000	15,000
Tables and clips made of stainless steel	100	80	8,000
Stainless steel tanks capacity of 1 m 3	4	500	2,000
tables with racks	2	2000	4,000
Fat separating machine	10	400	4,000
cooling tanks of Stainless steel, capacity 2 tons	2	4000	8,000
مبرد صفائحي	2	2500	5,000
Pipes	1	3000	3,000
Generator	1	1000	1,000
Special equipments for the cooling room	1	5000	5,000
Others (water tanks, pumps, tools. etc.)	1	3000	3,000
Oven	1	3000	3,000
Vacuum packaging machine	1	2000	2,000
Lab tools	1	400	400
Pasteurization device	1	4000	4,000
Other tools	1	-	7,740
Total (JD)			85,140

Table 29 Furniture Capital Expenditure

Office Furniture	2	1,000	2,000
Laboratory equipment	1	1,000	1,000
Total			3,000

Table 30 Transporation and Vehicles Capex

Rrefrigerated vehicles	2	15,000	30,000
Medium vehicle (pick-up)	1	15,000	15,000
Total			45,000

Table 31 Pre Operating Expenses

Governmental fees	1,000
Transportation	1,500
Marketing	2,000
Training	1,000
Total	5,500

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 32 Working Capital

Operating expenses	3	%25	90,560
General and administrative expenses	3	%25	712
Marketing expenses	3	%25	1,231
Indirect Salaries	3	%25	13,000
Total			105,503

Table 33 Expenses Summary

Land	50,000	10%	55,000	%13.7
Construction works	71,650	10%	78,815	%19.6
Machinery & Equipment	85,140	10%	93,654	%23.3
Furniture	3,000	10%	3,300	%0.8
Vehicles	45,000	10%	49,500	%12.3
Pre-Operating Expenses	5,500	10%	6,050	%1.5
Working capital	105,503	10%	116,054	%28.8
Total (JD)	365,793		402,373	100%

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (402,373) JD, where 50% of the project will be financed through loans (201,186) JD, and the remaining (50%) through (Equity) with (201,186) JD. The following table summarizes the general structure of the required financing:

Table 34 Sources of Funding

Equity (Self Financing)	201,186	%50
Loans	201,186	%50
Total	402,373	%100
Capital Expenditures	280,269	%69.65
Pre-Operating Expenses	6,050	%1.50
Working Capital	116,054	%28.84
Total	402,373	%100.00

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 35 Operating Expenses

Raw Materials	254,352	508,704	518,878	529,256	539,841	550,638	561,650	572,883	584,341	596,028
Utilities	9,848	20,091	20,492	20,902	21,320	21,747	22,182	22,625	23,078	23,539
Maintenance	0	1,005	1,025	1,045	1,066	1,087	1,109	1,131	1,154	1,177
Disposables	2,462	5,023	5,123	5,226	5,330	5,437	5,545	5,656	5,769	5,885
Direct Salaries	246	502	512	523	533	544	555	566	577	588
Others	62,400	64,272	66,200	68,186	70,232	72,339	74,509	76,744	79,046	81,418
Total	32,931	59,960	61,223	62,514	63,832	65,179	66,555	67,961	69,397	70,864

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 36 General and Administrative Expenses

Telecom	500	510	520	531	541	552	563	574	586	598
Hospitality	250	255	260	265	271	276	282	287	293	299
Stationary and Printing	200	204	208	212	216	221	225	230	234	239

Consulting	1,000	1,020	1,040	1,061	1,082	1,104	1,126	1,149	1,172	1,195
Training	500	510	520	531	541	552	563	574	586	598
Insurance	140	143	146	149	152	155	158	161	164	167
Miscellaneous	259	264	269	275	280	286	292	298	303	310
Total	2,849	2,906	2,964	3,024	3,084	3,146	3,209	3,273	3,338	3,405

Marketing Expenses

Table 37 Marketing Expenses

Marketing Expenses	4,924	10,045	10,246	10,451	10,660	10,873	11,091	11,313	11,539	11,770
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Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 38 Expected monthly salaries (2018-2027)

Indirect staff

Project Manager	1,000	1,030	1,061	1,093	1,126	1,159	1,194	1,230	1,267	1,305
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Agricultural engineer	800	824	849	874	900	927	955	984	1,013	1,044
Food processing technician	400	412	424	437	450	464	478	492	507	522
Marketing and supply officer	350	361	371	382	394	406	418	430	443	457
Security	300	309	318	328	338	348	358	369	380	391

Direct staff

Maintenance technician	400	412	424	437	450	464	478	492	507	522
Laboratory technician	400	412	424	437	450	464	478	492	507	522
Technical production supervisor	350	361	371	382	394	406	418	430	443	457
Workers and drivers	300	309	318	328	338	348	358	369	380	391
Cleaners	300	309	318	328	338	348	358	369	380	391

Table 39 Annual Expected Salaries

Indirect staff

Project Manager	16,000	16,480	16,974	17,484	18,008	18,548	19,105	19,678	20,268	20,876
Agricultural engineer	12,800	13,184	13,580	13,987	14,407	14,839	15,284	15,742	16,215	16,701
Food processing technician	12,800	13,184	13,580	13,987	14,407	14,839	15,284	15,742	16,215	16,701
Marketing and supply officer	5,600	5,768	5,941	6,119	6,303	6,492	6,687	6,887	7,094	7,307
Security	4,800	4,944	5,092	5,245	5,402	5,565	5,731	5,903	6,080	6,263
Total Non-Direct staff	52,000	53,560	55,167	56,822	58,526	60,282	62,091	63,953	65,872	67,848

Direct staff

Maintenance technician	6,400	6,592	6,790	6,993	7,203	7,419	7,642	7,871	8,107	8,351
Laboratory technician	6,400	6,592	6,790	6,993	7,203	7,419	7,642	7,871	8,107	8,351
Technical production supervisor	11,200	11,536	11,882	12,239	12,606	12,984	13,373	13,775	14,188	14,613
Workers and drivers	28,800	29,664	30,554	31,471	32,415	33,387	34,389	35,420	36,483	37,577
Cleaners	9,600	9,888	10,185	10,490	10,805	11,129	11,463	11,807	12,161	12,526
Total of direct staff	62,400	64,272	66,200	68,186	70,232	72,339	74,509	76,744	79,046	81,418
Grand Total	114,400	117,832	121,367	125,008	128,758	132,621	136,600	140,698	144,918	149,266

Depreciations

The following tables illustrate cost of Capex and Depreciation rates:

Table 40 Capex and Depreciation Expenses

Land	55,000	%0.0	%0.0
Construction Works	78,815	%5.0	%0.0
Machineries	93,654	%10.0	%2.0
Furniture	3,300	%10.0	%5.0
Vehicles	49,500	%10.0	%0.0
	280,269		

Table 41 Depreciations and Additions on Construction Works

Total Fixed Assets	280,269	282,307	284,391	286,522	288,700	290,928	293,207	295,537	297,921	300,360
Total Depreciation Expenses	18,586	18,790	18,998	19,211	19,429	19,652	19,880	20,113	20,351	20,595
Total Accumulated Depreciation	18,586	37,376	56,374	75,586	95,015	114,667	134,547	154,660	175,012	195,607
Total Additions	0	2,038	2,084	2,131	2,179	2,228	2,279	2,331	2,384	2,438
Total Net Book Values	261,683	244,931	228,016	210,936	193,685	176,261	158,660	140,877	122,910	104,753

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 42 Loan Details

Annual Interest Rate	9.50%
Loans Period	5
Loan Starts at year:	2018

2018										201,186
2019		52,396		19,113		33,284				167,903

2020	52,396	15,951	36,445	131,457
2021	52,396	12,488	39,908	91,549
2022	52,396	8,697	43,699	47,850
2023	52,396	4,546	47,850	0

Income Statement

Table 43 Income Statement

Revenues										
Sales Revenues	492,415	1,004,526	1,024,617	1,045,109	1,066,011	1,087,332	1,109,078	1,131,260	1,153,885	1,176,963
Gross Operating Revenues	492,415	1,004,526	1,024,617	1,045,109	1,066,011	1,087,332	1,109,078	1,131,260	1,153,885	1,176,963
Operating Expenses	(362,239)	(659,556)	(673,454)	(687,651)	(702,154)	(716,970)	(732,105)	(747,566)	(763,362)	(779,499)
Gross Operating Profit	130,175	344,971	351,163	357,458	363,857	370,362	376,973	383,693	390,523	397,464
Gross Profit Percentage	%26	%34	%34	%34	%34	%34	%34	%34	%34	%34
Salaries and Benefits (Indirect Staff)	(52,000)	(53,560)	(55,167)	(56,822)	(58,526)	(60,282)	(62,091)	(63,953)	(65,872)	(67,848)
General and Administrative Expenses	(2,849)	(2,906)	(2,964)	(3,024)	(3,084)	(3,146)	(3,209)	(3,273)	(3,338)	(3,405)

Marketing Expenses	(4,924)	(10,045)	(10,246)	(10,451)	(10,660)	(10,873)	(11,091)	(11,313)	(11,539)	(11,770)
Pre Operating Expenses	(6,050)									
Gross Indirect Expenses	(65,823)	(66,511)	(68,377)	(70,296)	(72,271)	(74,301)	(76,390)	(78,539)	(80,749)	(83,023)
Income before Interest, Depreciation and Tax	64,352	278,459	282,786	287,162	291,587	296,061	300,583	305,155	309,774	314,441
Fixed Assets Depreciations	(18,586)	(18,790)	(18,998)	(19,211)	(19,429)	(19,652)	(19,880)	(20,113)	(20,351)	(20,595)
Income before Tax and Interests	45,766	259,669	263,788	267,950	272,158	276,409	280,703	285,042	289,423	293,846
Bank Interests	(19,113)	(15,951)	(12,488)	(8,697)	(4,546)	0	0	0	0	0
Income Before Tax	26,653	243,719	251,299	259,253	267,612	276,409	280,703	285,042	289,423	293,846
Net Profit	26,653	243,719	251,299	259,253	267,612	276,409	280,703	285,042	289,423	293,846
<i>Net Profit Percentage</i>	<i>%5</i>	<i>%24</i>	<i>%25</i>	<i>%25</i>	<i>%25</i>	<i>%25</i>	<i>%25</i>	<i>%25</i>	<i>%25</i>	<i>%25</i>
Compulsory Reserves	(2,665)	(24,372)	(25,130)	(25,925)	(26,761)	(27,641)	(28,070)	(28,504)	(28,942)	(29,385)
Retained Earnings	23,988	243,335	469,504	702,832	943,682	1,192,450	1,445,083	1,701,621	1,962,101	2,226,562

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 44 Expected Cashflows

Cash flows from operating activities

Net profit (loss)	26,653	243,719	251,299	259,253	267,612	276,409	280,703	285,042	289,423	293,846
Non-monetary adjustments										
Bank benefits	19,113	15,951	12,488	8,697	4,546	0	0	0	0	0
Consumption	18,586	18,790	18,998	19,211	19,429	19,652	19,880	20,113	20,351	20,595
Operating cash flows before changes in working capital	64,352	278,459	282,786	287,162	291,587	296,061	300,583	305,155	309,774	314,441
Shatrage (increase) in inventory	(21,196)	(21,196)	(848)	(865)	(882)	(900)	(918)	(936)	(955)	(974)
Decrease (increase) in receivables	(82,069)	(85,352)	(3,348)	(3,415)	(3,484)	(3,553)	(3,624)	(3,697)	(3,771)	(3,846)
Change in working capital	(103,265)	(106,548)	(4,196)	(4,280)	(4,366)	(4,453)	(4,542)	(4,633)	(4,726)	(4,820)
Net cash flows (used) from operating activities	(38,913)	171,911	278,590	282,882	287,221	291,608	296,041	300,522	305,048	309,621
Cash flows from investing activities										
(Purchase) of fixed assets	(280,269)	(2,038)	(2,084)	(2,131)	(2,179)	(2,228)	(2,279)	(2,331)	(2,384)	(2,438)
Net flows (used) from investment activities	(280,269)	(2,038)	(2,084)	(2,131)	(2,179)	(2,228)	(2,279)	(2,331)	(2,384)	(2,438)
Cash flows from financing activities										
Equity capital	201,186									
Loan Amortization	(33,284)	(36,445)	(39,908)	(43,699)	(47,850)	0	0	0	0	0
Bank Interest Rate	(19,113)	(15,951)	(12,488)	(8,697)	(4,546)	0	0	0	0	0
Loans	201,186	0	0	0	0	0	0	0	0	0

Net cash flows (used) from financing activities	349,976	(52,396)	(52,396)	(52,396)	(52,396)	0	0	0	0	0
Net increase (decrease) in cash	30,794	117,477	224,110	228,355	232,646	289,380	293,763	298,191	302,664	307,183
Cashflows at the Beginning of Period	0	30,794	148,271	372,381	600,736	833,382	1,122,761	1,416,524	1,714,715	2,017,380
Cashflows at the End of Period	30,794	148,271	372,381	600,736	833,382	1,122,761	1,416,524	1,714,715	2,017,380	2,324,562

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 45 Expected Balance Sheet

Assets

Current assets

Fund	30,794	148,271	372,381	600,736	833,382	1,122,761	1,416,524	1,714,715	2,017,380	2,324,562
Inventory	21,196	42,392	43,240	44,105	44,987	45,886	46,804	47,740	48,695	49,669
Accounts receivable	82,069	167,421	170,769	174,185	177,669	181,222	184,846	188,543	192,314	196,160
Total assets traded	134,059	358,084	586,390	819,025	1,056,037	1,349,870	1,648,175	1,950,999	2,258,389	2,570,392

Non-current assets

Fixed assets (net)	261,683	244,931	228,016	210,936	193,685	176,261	158,660	140,877	122,910	104,753
Total non-current assets	261,683	244,931	228,016	210,936	193,685	176,261	158,660	140,877	122,910	104,753
Total assets	395,742	603,015	814,407	1,029,961	1,249,722	1,526,131	1,806,834	2,091,876	2,381,299	2,675,144
Liabilities										
Current liabilities										
Remaining amount of Loan	36,445	39,908	43,699	47,850	0	0	0	0	0	0
Total current liabilities	36,445	39,908	43,699	47,850	0	0	0	0	0	0
Non-current liabilities										
long term loans	131,457	91,549	47,850	0	0	0	0	0	0	0
Total long - term liabilities	131,457	91,549	47,850	0	0	0	0	0	0	0
Total liabilities	167,903	131,457	91,549	47,850	0	0	0	0	0	0
Property rights										
Member Contributions	201,186	201,186	201,186	201,186	201,186	201,186	201,186	201,186	201,186	201,186
Compulsory reserve	2,665	27,037	52,167	78,092	104,854	132,494	160,565	189,069	218,011	247,396
Profit (loss) round	23,988	243,335	469,504	702,832	943,682	1,192,450	1,445,083	1,701,621	1,962,101	2,226,562
Total equity	227,840	471,558	722,857	982,111	1,249,722	1,526,131	1,806,834	2,091,876	2,381,299	2,675,144
Total liabilities and equity	395,742	603,015	814,407	1,029,961	1,249,722	1,526,131	1,806,834	2,091,876	2,381,299	2,675,144

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 46 Free Net Cash flows Table

Net Free Cashflows	(402,373)	(32,863)	169,873	276,506	280,751	285,042	289,380	293,763	298,191	302,664	307,183	4,010,921
Discount Factor	1.00	0.89	0.80	0.71	0.64	0.57	0.51	0.45	0.40	0.36	0.32	0.32
Net Present Value for Cash	(402,373)	(29,351)	135,507	196,996	178,646	161,994	146,884	133,174	120,736	109,451	99,214	1,295,452

Table below (47) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 47 payback period

Net Free Cashflows	(402,373)	(32,863)	169,873	276,506	280,751	285,042	289,380	293,763	298,191	302,664	307,183
Project Value Returned	402,373	435,236	265,362	(11,144)	(291,895)	(576,937)	(866,317)	(1,160,079)	(1,458,270)	)	(2,068,117)
Payback Period (Year)	2	1	1	0	0	0	0	0	0	0	0

Table 48 Financial Analysis Results

Weighted Average Cost of Capital (WACC)	%11.97
Net Present Value for Cashflows	2,146,330
Payback Period	2
Internal Rate of Return	%45.53

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 394,714 Jordan Dinars considering that the project provides 7 Job Opportunities for the governorate residents.

Table 49 Sensitivity Analysis

Original Scenario	45.53%	2	11.97%
Revenues declined by 10%	32.16%	4	11.97%
Operating Expenses Increased by 10%	35.30%	3	11.97%